

The Rakia mission was an Israeli space mission conducted in April 2022 as part of Ax-1 (Axiom Mission 1) , the first all-private crewed mission to the International Space Station (ISS). The mission featured Eytan Stibbe, a former fighter pilot and impact investor, who became the second Israeli astronaut and the first to visit the ISS. Managed by the Rakia team supported by the Ramon foundation and in collaboration with the Israel Space Agency (ISA) under the Ministry of Innovation, Science, and Technology, and the Ministry of Education, the mission was designed to advance Israeli science, education, and art through microgravity research and public engagement.

Overview

The name "Rakia" (Hebrew: רַקִּיָּא), meaning "sky" or "firmament," was selected through a public vote in 2021. Stibbe served as a mission specialist alongside Commander Michael López-Alegría, investor Larry Connor, and philanthropist Mark Pathy. The crew launched on April 8, 2022, aboard a SpaceX Crew Dragon Endeavour and spent 17 days in orbit, extending beyond the original 10-day schedule due to weather delays for the return splashdown. The Ax-1 team worked onboard the ISS together with Expedition 67 that included SpaceX Crew 3 (NASA astronauts: ISS commander Thomas Marshburn, Raja Chari and Kayla Barron and ESA astronaut Matthias Maurer) and Soyuz MS-21 (Roscosmos astronauts Oleg Artemyev, Denis Matveev and Sergey Korsakov)

Science and technology

A central pillar of the Rakia mission was the execution of dozens of scientific experiments and technological demonstrations. A science committee that included leading figures in Israeli academia, led by Inbal Kreiss, selected projects from diverse fields such as medicine, astrophysics, and agriculture.

Fluidic Telescope (FLUTE): The Technion in collaboration with NASA's Ames Research Center to test the production of high-quality optical lenses in microgravity by shaping liquid polymers. This experiment demonstrated the potential for manufacturing large-scale telescopes in space.

ILAN-ES: An atmospheric research project focused on observing Transient Luminous Events (TLEs) that are short-lived, high-altitude electrical breakdown phenomena that occur far above normal thunderstorm clouds and lightning from space. The experiment builds on the legacy of the MEIDEX project conducted by Ilan Ramon aboard Space Shuttle Columbia (STS-107) in 2003. Following its success on Ax-1, ILAN-ES became a recurring payload on subsequent missions, including Ax-2 and Ax-3. For the Ax-4 mission, the project evolved into the UHU experiment, an international collaboration with the Hungarian Space Program (HUNOR).

Brain.space: A study of electrical brain activity (EEG) to monitor cognitive function and neural changes in astronauts, providing data on how the brain adapts to microgravity.

CRISPR Diagnosis: A proof-of-concept for using gene-editing technology for the rapid detection of pathogens in space, essential for long-term Lunar or other deep space missions.

AstroRad Radiation Vest: A test of the StemRad-developed protective suit designed to shield astronauts from cosmic radiation. Stibbe wore the vest during daily activities to evaluate its ergonomics and comfort for future space missions. The vest was successfully tested on the lunar Artemis 1 mission around the moon, in November 2022.

Flexible Antenna (NSLComm): A technological demonstration of a deployable, fabric-based antenna designed to enable high-bandwidth communications for nano-satellites. The experiment successfully validated the antenna's deployment mechanism in microgravity.

Education and art

Under the vision of "Space for All," the mission utilized the ISS as a classroom to inspire the next generation of scientists and engineers. Educational content was created and broadcast in

Hebrew, Arabic, and English, reaching hundreds of thousands of students. Key projects included "A Bedtime Story from Space," where Stibbe read children's books to Hebrew and Arabic-speaking audiences.

The Rakia mission featured a dedicated art component curated by Udi Edelman and Hadas Gur-Arie. Israeli artists contributed 15 works inspired by space exploration and designed to function in or react to microgravity, exploring the intersection of human creativity and the space environment.

Mission control and outreach

During the mission, a dedicated control room and a visitor center were established at the offices of Check Point Software Technologies in Tel Aviv. This facility served as the primary communication hub between the Israeli ground team and Stibbe on the ISS. The visitor center hosted over 20,000 visitors during the mission, including scientists, artists, students and families, offering immersive exhibits on the mission's activities and goals and giving an overview about the daily life of astronauts while in space.

Current activities and legacy

Following the mission's conclusion, Rakia transitioned into a Public Benefit Company (PBC) focused on the long-term development of the Israeli space ecosystem. The organization facilitates access to microgravity research for entrepreneurs and researchers while aligning Israeli technology with international exploration goals, including the Artemis program.

Israel Space Forum: Rakia established this collaborative platform to unite space-tech companies, government agencies, and academic researchers. The forum aims to accelerate technological growth within the global NewSpace economy and coordinate Israeli contributions to future lunar and Martian missions.

Space Campus IL: The organization leads Israel's first student space community, which connects university students with industry professionals. This initiative focuses on developing the next generation of space professionals through academic collaborations and specialized training programs.

Cultural and Artistic Initiatives: Rakia continues to manage a cultural branch that grew from the Ax-1 mission. This includes curating art exhibitions designed for space, hosting conferences on the intersection of humanity and technology, and promoting "Space for All" through public outreach.

Educational Leadership: The mission's long-term impact on global space education was highlighted in 2025 with the appointment of Eytan Stibbe as Chancellor of the International Space University (ISU).

International collaborations: During Axiom Mission 4 (Ax-4), Rakia collaborated on international educational and space-related initiatives connected to the mission and its participating astronauts, including Indian astronaut Shubhanshu Shukla. As part of the mission, Rakia and the HUNOR Hungarian Space Program also collaborated on the UHU (Upper Atmosphere – Hungarian Universe) experiment conducted aboard the International Space Station.

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